Conditioned Anxiety Mechanism as a Basis for a Procedure of Control Module of an Autonomous Robot

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Abstract. This paper is devoted to the problem of self-control of autonomous robot in a complex, unknown environment. In such an environment it is impossible to predict all situations the robot could be faced with. Because of this it is necessary to equip the robot with control procedures that allow it to avoid dangerous scenarios. Mechanisms that serve to avoid threatening events have been worked out during evolution and living organisms are equipped with them. Conditioned anxiety is one of such mechanisms. In this paper the way in which this mechanism can be adapted to control of behaviour of autonomous robot, is presented. The effectiveness of the proposed approach has been verified by using V-REP simulator.

Keywords: Emotional robot · Autonomous agent · Conditioned anxiety · Artificial intelligence · Learning · Control

1 Introduction

Artificial intelligence (AI for abbreviation) systems are rooted in various originals. Most of these originals are connected with human cognitive abilities. AI systems based on various types of logic, fuzzy inference, syntactic algorithms, semantic nets and mathematical linguistics can be put as examples [12]. Furthermore, artificial neural networks (ANNs for abbreviation) are based on structures

© Springer International Publishing AG 2017
L. Rutkowski et al. (Eds.): ICAISC 2017, Part II, LNAI 10246, pp. 390–398, 2017.
DOI: 10.1007/978-3-319-59060-8_35

and functions of biological neural systems [26,31,35]. The aforementioned AI systems are universal ones in the context of solving a wide spectrum of problems. They are, however, time consuming and therefore possibilities of applying them to on-line problems are limited.

The idea of autonomous agent is the one in which functional possibilities of AI systems are used widely. On the one hand an agent is, usually, equipped with cognitive modules which enable the agent to perceive and understand the environment in which it acts [5–9,11,24,25] and [12], Chap. 14. On the other hand, however, as it has been aforementioned, such types of systems are insufficient for solving on-line problems connected with the agent self-control. Therefore, a crucial stream of studies in cybernetics and robotics consists in searching effective mechanisms of on-line control. Biocybernetics, in particular neurocybernetics, is one of the areas of such studies [27–30] the more so because biological models are one of the crucial basis of cybernetics as a whole [34].

Autonomous robots as well as biological individuals are examples of embodied autonomous agents and can be studied in the frame of autonomous systems theory [4,21]. They often operate in complex, unknown environments in which they can be faced with dangerous situations. Space missions, including exploration of Mars by using remote-controlled or autonomous vehicles, are examples of such risky ventures [2,14,15,18,33]. Therefore, there is a great demand for control procedures that enable to avoid by a robot situations that are potentially dangerous. Conditioned anxiety which has been created by evolution process is one of such mechanism [13,16,17,19,23]. It can be modelled and implemented in a robot. Such approach is situated in the studies concerning so called emotional agents in which models of emotions are important component of a decision making process [10,22,32].

In this paper a proposal of adaptation of the conditioned anxiety mechanism to control of autonomous robot is described. This mechanism enable the robot to learn quickly the circumstances of the threatening situation it has been faced with. As a consequence the robot will avoid in future similar circumstances because there is significant probability that they are correlated with the experienced danger. It should be mentioned that this paper is a continuation of the studies that concern scene analysis and self-control of autonomous agents operating in complex environment [5–9], in particular, the investigations concerning adaptation of conditioned mechanisms to AI systems [3].

This paper is organized in the following way. In the next section justification significance for carrying out this studies is presented. The way of adaptation of the mechanism of conditioned anxiety to control of autonomous robot is described in Sect. 3, whereas the tests of the proposed approach efficiency are described in Sect. 4.

2 Motivations

As it has been aforementioned, evolution worked out conditioned reactions that, among others, enables the living individuals to learn a proper reaction quickly in

critical situations [17]. Such types of reactions are usually connected with strong emotions or, frequently, are triggered by them [17]. Autonomous robots lack of such mechanisms. Although emotions are tried to be modelled and implemented in robots as an element of a decision making algorithms - see [1,10,22,32] - the scheme which consists in modelling conditioned reactions and implementing them as computer and robotic algorithms seems to be a pioneering approach [3]. It seems, furthermore, that conditioned reactions have not been neither modelled cybernetically nor implemented as algorithms in connection with emotions.

Most of contemporary robots are remote-controlled. Such solution has, however, crucial drawbacks: communication can be broken especially in critical conditions, for instance where meteorological conditions are hard, during military operations or in space exploration. Furthermore, communication channels can suffer from low-bandwidth or high-latency, especially during interplanetary communication. Therefore, there is a great demand of high degree of autonomy for spacecrafts. This problem was articulated explicitly: "*The vehicles used to explore the Martian surface require a high degree of autonomy to navigate challenging and unknown terrain, investigate targets, and detect scientific events. Increased autonomy will be critical to the success of future missions.*" [2]. The Mars is explored intensively by using wheeled unmanned vehicles - until now three generates of vehicles can be constructed: Sojourner vehicle used during Pathfinder mission in 1997, Spirit and Opportunity - Mars Exploration Rover vehicles used during the mission in 2004 and Mars Science Laboratory vehicle used during the mission in 2009 [2]. Since it is impossible to predict all threatening events during a mission, the demand for control procedures that enables effective learning of proper reactions in critical situations is significant. On the other hand, studies concerning this topic are at the very initial stage. This paper is intended to fill partially this gap.

3 The Proposed Algorithm

In this section the proposed algorithm is described. In the first subsection the very idea rooted in neuropsychology is presented. In the next subsection the algorithm is proposed.

3.1 Conditioned Anxiety as a Mechanism of Control

As it has been aforementioned agents which act in a complex and unknown environment are not able to predict all situations they can be faced with. Therefore they should be equipped with mechanisms protecting them from dangers. Such mechanisms as conditioned aversion, conditioned anxiety and conditioned fear plays great role in evolution as a protecting mechanisms [13,16,17,19,23]. Let us consider their structure.

A biological individual confronted by a threatening event should avoid it in future. Such events are experienced together with some accompanying circumstances. Since there is probable that these circumstances are not accidental but,

at least some of them coincide with the experienced danger, the strategy consists in avoiding them in future seems to be a good solution which should enable the individual to avoid the experienced danger again. Avoiding circumstances that proceeded the dangerous event is evidently even more efficient solution. The learning process which enables avoiding the dangerous events should be very quick because each such event can cause the death of the individual. Therefore, single experience of this type should be sufficient to learn effectively the proper schema of reaction. Conditioned anxiety is such a biological mechanism. If an animal was attacked by a predator in a strongly limited terrain, for instance in a cramped ravine, it should avoid cramped places in future provided that it managed to survive itself in confrontation with the attacker. Similarly, the animal which survived an attack of predatory bird on the field should avoid in future open spaces. Since, as it has been aforementioned, the learning process should be quick and effective, the victim associates the proceeding circumstances with the attack by generate the strong anxiety conditioned by these circumstances. In such a way claustrophobia or fear of open spaces is generated. On the one hand such mechanism is experienced by biological individuals as a great discomfort, especially if they have neurotic character. On the other hand, however, it is very effective as a source of a protective strategy that consists in steering clear of potentially dangerous places and situations.

3.2 The Proposed Algorithm

The conditioned anxiety generated in dangerous situation can be implemented in artificial systems in the following way. First of all let us remark that, from the cybernetic point of view, a mobile self-controlling robot has to be organized as a cybernetic autonomous system. The basis of the theory of such systems has been founded - see [20,21] - and is developed [4]. Thus, let us assume that the correlator, i.e. the module in which the knowledge of the autonomous system is stored and processed, is equipped with an array in which data registered by the robot receptors during a declared period of time, for instance last thirty seconds, are written and updated, let us say, every second. This array corresponds to short-term memory of biological individuals. If the robot got into dangerous situation, which can be recognized by it as threatening because of partial damage of the vehicle or serious problems with homeostasis or navigation after coming onto extremely difficult terrain, then the content of the short-term memory is written to the list of the arrays that encode the circumstances associated with the threatening events, one array for one event. This list corresponds to triggers that cause conditioned anxiety in biological systems. During the robot mission the list should be continuously compared to the contain of short-term memory in order to detect circumstances which are encoded as associated with a danger. If the current circumstances encoded in short-term memory are recognized as identical with one of the patterns written in the list, then the robot reversed itself as quickly as possible in order to change the circumstances. Since the procedure should act extremely quickly, the list should be implemented at the level of programmable electronics. Thus, the outline of the algorithm of the creating

the mechanism which is analogous to conditioned anxiety can be written in the following way:

- analyze continuously the readings of the receptors;
- analyze continuously the readings of the detectors which track the inner state of the robot (let us call them: inner detectors);
- **if** readings from the inner detectors are alarming **then** *create* the new array in the list of arrays which encode the circumstances associated with the threatening events;
- *write* the readings from the receptors for the last time interval created array;

In the above notation of the algorithm, the crucial actions are written in italics. The *time interval* denotes the aforementioned declared period of time which defines uploading of the short-term memory.

The implemented conditioned anxiety-like mechanism acts in the following way (the outline):

- browse continuously the list of arrays which encode the circumstances associated with the threatening events;
- **if** the content of the short-term memory is similar to the content of one of the browsed arrays **then** immediately start to reverse;
- continue the reversion **until** the content of the short-term memory is similar sufficiently to the content of the array;

4 Experiments and Results

The algorithm proposed in the previous section has been tested by using V-REP (Virtual Robot Experimentation Platform) simulator.[1] The algorithm was implemented by using Robot Operating System (ROS).[2] The scene on which the robot operated had been created - see Fig. 1. Two types of obstacles were created on the scene. Big obstacles, which were destroying for the robot if it collides with them and the small ones, denoted on the scene as thin cubicoids with squares bases, which damaged the robot if it run onto it. It was assumed that the robot could be damaged two times by the cubicoids but the third time was destroying. It was also assumed that a damaged robot could continue its mission. Exploration of the scene in order to create its map was the task of the robot. A mission lasted 150 s. If the robot was not destroyed during performing the task, then the mission was regarded as successful. An example of the map created by the robot during a successful mission is shown in Fig. 2.

The robot, simulated as a three-wheeled vehicle, was equipped with two senses. The first one, the LIDAR (Light Detection and Ranging) can detect only big obstacles. The robot was equipped with the algorithm of navigation which allowed it to avoid collisions with big obstacles. Ultrasonic detector, fastened to

[1] http://www.coppeliarobotics.com.
[2] http://wiki.ros.org/.

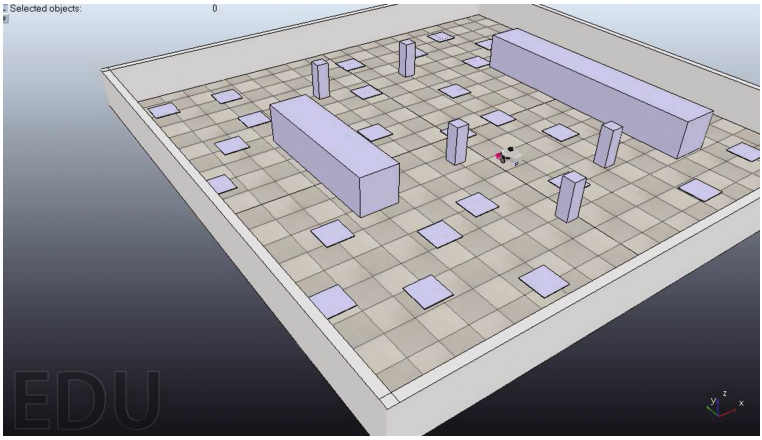


Fig. 1. The scene.

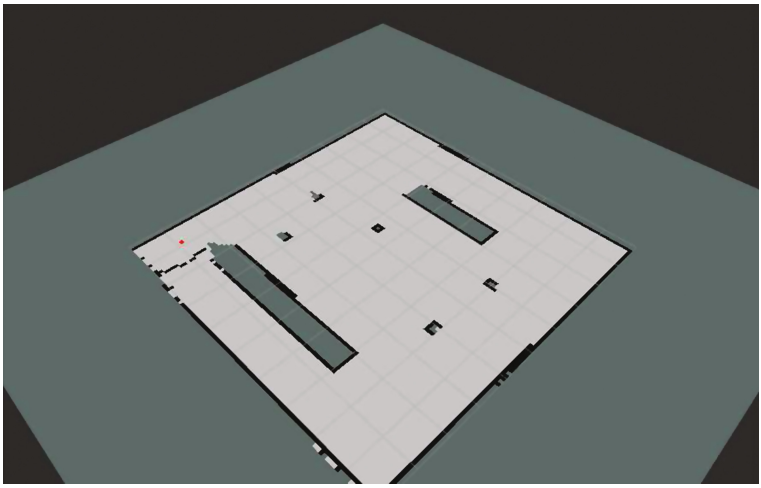


Fig. 2. The map of the environment - an obstacle floor map - created by the robot as a result of its mission - an example of a result of a successful mission.

the extension arm on the front of the robot was directed down and it can detect only cubicoids - see Fig. 3. The robot, however, was not a priori equipped with the knowledge that running onto cubicoids caused by its damage. The cognitive module of the robot had to associate detection of the cubicoid with the fact that the vehicle would be damaged a few seconds after a cubicoid detection. This association was created by using the algorithm described in the previous section. As it has been aforementioned, creating of the map of the scene was the task of the robot. The maps were created by using Simultaneous Localization and Mapping (SLAM) technique, realized by gmapping node available in ROS.

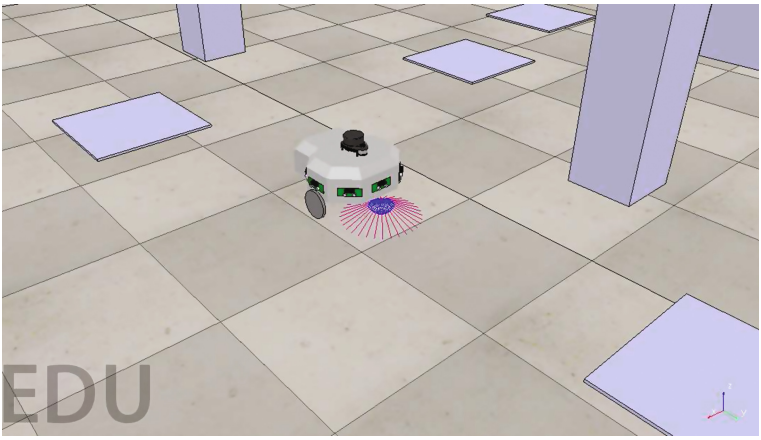


Fig. 3. The robot operating on the scene.

This node enables creation a 2-D occupancy grid map, for instance a building floor plan. The tests were performed both for the robot without conditioned anxiety, let us call it as the simple robot, and for the robot with the model of conditioned anxiety implemented in the way described in the previous section - let us call this type of robot as the emotional one. In each test the robot started at a randomly chosen point of the scene presented in Fig. 1. For twelve tests for the simple robot seven were unsuccessful. This means that 58% of tests ended as failures. For eleven tests for the emotional robot only three were failures which is equal to 27% of all tests.

5 Concluding Remarks

In this paper the way in which the mechanism of conditioned anxiety can be adapted to control of behaviour of autonomous robot is presented. The performed tests shown that the mechanism is effective. Probability of failure for the simple robot was over two times greater than for the emotional one. It should be stressed that the algorithm has been implemented in the extremely simple version. Therefore, there were no problems with associate the circumstances with the threatening event. Furthermore, the created scene was very simple as well. Thus, the obtained results should be regarded as preliminary.

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